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Integrating multi-omics data reveals the antitumor role and clinical benefits of gamma-delta T cells in triple-negative breast cancer

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Abstract

Background Gamma-delta ($\gamma\delta$) T cells are a critical component of the tumor microenvironment and have been recognized as a promising biomarker and target for cancer therapy. Increasing evidence suggests that $\gamma\delta$ T cells play distinct roles in different cancers. However, the impact of $\gamma\delta$ T cells in breast cancer remains controversial.

Methods In this study, we investigated the role of $\gamma\delta$ T cells in breast cancer using a comprehensive approach, including bulk and single-cell sequencing, radiomics based on magnetic resonance imaging (MRI), genomic data, and immunohistochemistry. Single-cell RNA profiling was used to infer the potential lineage evolution of $\gamma\delta$ T cells and their interactions with other immune cells. Bulk RNA sequencing was included to uncover the heterogeneity in signaling pathways, as well as radiotherapy and immunotherapy responses, among patients with varying levels of $\gamma\delta$ T cell abundance. Genomic analysis was used to recognize the critical gene mutations with the infiltration of $\gamma\delta$ T cells. Immunohistochemistry was performed to validate the prognostic value of $\gamma\delta$ T cells in breast cancer patients. Lastly, radiomics was used to establish a correlation between the abundance of $\gamma\delta$ T cells and the features of MRI images.

Results The $\gamma\delta$ T cell infiltration was closely associated with favorable prognosis in triple-negative breast cancer (TNBC) but not in other subtypes of breast cancer. $\gamma\delta$ T cells may exert antitumor effects through intrinsic lineage evolution or interact with antigen-presenting cells through ligand-receptor pairs. Patients with a high $\gamma\delta$ T cell abundance may benefit more from chemotherapy or radiotherapy alone than their combination. Additionally, patients with a high $\gamma\delta$ T cell abundance were more likely to benefit from immunotherapy. Finally, we established a radiomic model based on dynamic contrast-enhanced-MRI, which indicated the potential for estimating the $\gamma\delta$ T cell abundance for patients with TNBC.

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Conclusion Our study provides novel insight and a theoretical basis for individualized therapy of patients with TNBC based on $\gamma\delta$ T cells.

Keywords Gamma-delta T cells, Triple-negative breast cancer, Immunotherapy, Tumor microenvironment, Radiomics

Introduction

Gamma-delta ($\gamma\delta$)T lymphocytes have garnered significant interest among oncologists recently. As a unique and vital subgroup of tumor-infiltrating lymphocytes, $\gamma\delta$ T cells are characterized by T cell receptors (TCR) composed of γ and δ chains, which differ from traditional $\alpha\beta$ T cells whose TCR consisted of α and β chains [1, 2] (CD4, or CD8 T cells). The unique structure of the TCR allows $\gamma\delta$ T cells to recognize antigens independently of major histocompatibility (MHC) molecular pathways, enabling a rapid immune response [1]. Additionally, $\gamma\delta$ T cells can secrete cytokines (interferon- γ , tumor necrosis factor- α) or express natural killer receptors (NKG2D) to exert cytotoxicity and antitumor effects [3]. However, evidence suggests a dual function of $\gamma\delta$ T cells in various cancers. For instance, Rong et al. [4] found that the imbalance of $\gamma\delta$ T cells may promote the progression of rectal cancer. A subgroup of $\gamma\delta$ T cells (interleukin (IL)-17 $\gamma\delta$ T cells) promotes ovarian cancer development through a complex interaction with small peritoneal macrophages [5]. Conversely, the prognostic benefit of $\gamma\delta$ T cells has been confirmed in several cancers, including cervical cancer, gastric cancer, and non-small lung cancer [6–8].

Breast cancer has been a leading cause of cancer-related death among women worldwide for several decades [9, 10]. Its high heterogeneity leads to distinct clinical outcomes for patients, particularly in triple-negative breast cancer (TNBC), which exhibits the most aggressive behaviors [11]. Apart from chemotherapy, limited treatment options restrict the overall survival and quality of life for patients with TNBC [12]. Therefore, developing novel treatments and risk stratification strategies is urgently needed to improve the prognosis of patients with TNBC. The emergence of single-cell sequencing and radiomics offers new strategies and hope for prognostic stratification and individualized cancer therapy for breast cancer patients [13–15]. Single-cell sequencing has a unique advantage in elucidating cell subpopulation heterogeneity and cell-cell interactions in the tumor microenvironment [16]. Radiomics provides an inexpensive and noninvasive evaluation strategy [17]. Through multi-omics exploration, the characteristics of the tumor microenvironment will be revealed more comprehensively and extensively.

Currently, the role of $\gamma\delta$ T cells in breast cancer remains controversial, and few studies have explored their role in breast cancer through multi-omics profiles. In this study, we employed multi-omics data, including bulk

transcriptome, single-cell ribonucleic acid (RNA) transcriptome, somatic mutational data, and radiomics dynamic contrast-enhanced magnetic resonance imaging (DCE-MRI), to investigate the potential role of $\gamma\delta$ T cells in the TNBC microenvironment. Our results confirmed that high $\gamma\delta$ T cell infiltration is associated with a favorable prognosis in patients with TNBC but not in patients with hormone receptor-positive or human epidermal growth factor receptor 2 (HER2). The $\gamma\delta$ T cell infiltration may induced by somatic gene mutations including Mucin 16 (*MUC16*), Dedicator Of Cytokines 11 (*DOCK11*), Laminin Subunit Alpha 2 (*LAMA2*), and Ryanodine Receptor 1 (*RYR1*). Notably, intrinsic lineage evolution may drive the direct cytotoxic effects of $\gamma\delta$ T cells.

Additionally, $\gamma\delta$ T cells may enhance the immune suppressive microenvironment by interacting with antigen-presenting cells (APCs) (conventional dendritic cells (cDCs) and macrophage). Most importantly, the $\gamma\delta$ T cell abundance could guide chemotherapy, radiotherapy, and immunotherapy. Finally, we found that the $\gamma\delta$ T cell abundance could be estimated through radiomics, which is significantly meaningful for individualized therapy in TNBC.

Methods

Data acquisition and preprocessing

Three single-cell RNA datasets were included in the study (Supplementary File: Table S1) from the gene expression omnibus database, including GSE176078 (10 primary TNBC), GSE180286 (2 primary TNBC), and GSE128223 (6 samples of purified human $\gamma\delta$ T cells from 3 donors). The multi-omics profiles of the cancer genome atlas breast invasive carcinoma datasets, including prognostic information, transcriptome, deoxyribonucleic acid copy number variation (CNV), and single nucleotide variation (SNV), were downloaded using the ‘TCGAbio-links’ package from The Cancer Genome Atlas (TCGA) [18]. The radiomics data from the TCGA-BRCA dataset were obtained from The Cancer Imaging Archive (TCIA) database (<https://dev.cancerimagingarchive.net/>). Additionally, the gene expression profiles and clinical characteristics of the METABRIC dataset were acquired from cBioPortal [19] (<https://www.cbioportal.org/>). The clinical characteristics of public datasets included in this study were presented in Supplementary File: Table S2.

Single-cell RNA profiles were analyzed using the Seurat (version 4.3.0) package. The same quality control standard was applied to the 12 TNBC samples integrated

from GSE180286 and GSE176078. And the same quality control procedures were applied to the 6 purified human $\gamma\delta$ T cell samples (GSE128223). Detailed quality control procedures are presented in Figures S1 and 2. The DoubletFinder [20] (version 2.0.3) and Harmony [21] (version 0.1.1) packages removed the batch effect and potential doublets, with the doublets ratio set as 0.8%. The SCTransform method was employed to normalize data and remove mitochondrial interference. The cluster packages were used to assist in selecting the resolutions of cell clusters. Nonlinear dimensionality reduction, including t-distributed stochastic neighbor embedding and uniform manifold approximation and projection, was used to visualize the similarity of cell expression profiles. Markers of cell annotation were obtained from online databases and previously published studies.

Patients and sample collection

A total of 61 TNBC specimens from patients who underwent surgery between February 2, 2016 and September 28, 2016, were retrospectively collected at Tianjin Medical University Cancer Institute and Hospital. Informed written consent was obtained from the participant. The study was approved by the Ethical Committee of Tianjin Medical University Cancer Institute and Hospital and adhered to the ethical guidelines of the Helsinki Declaration (ID: Ek2021021).

Immunohistochemistry

The immunohistochemistry (IHC) process was performed according to the manufacturers, including necessary positive and negative controls. With the EnVision two-step procedure, the IHC score was conducted on paraffin-embedded, 4 μ m sections, which were performed antigen retrieval method and incubated with TCR γ/δ monoclonal antibody TCR1061 (clone 5A6.E9, Thermo Fisher, USA) at 4 °C overnight, then rewarmed to room temperature. $\gamma\delta$ +T cell infiltrated into tumors was assessed by manually counting the number of positive cells in five high-magnification fields (40 $\times$) for each sample to obtain quantitative results. Two independent pathologists evaluated the number of $\gamma\delta$ +T cells. The more details of experiments were performed as previously described [6]. The groups were divided into high- and low- $\gamma\delta$ +T cell based on the median value of cell numbers. The Log-rank test was used to evaluate the statistical significance of survival probability across distinct groups.

Downstream analyses of scRNA-seq

The 'Vector' package [22] was an effective tool for unsupervised inference of developmental trajectories. The developmental directions of $\gamma\delta$ T cells were inferred using the 'Vector' package with default parameter. The

Monocle [23] (version 2.26.0) package was employed to analyze the expression profile evolution of $\gamma\delta$ T cell trajectories and to identify distinct cell states of $\gamma\delta$ T cells. The 'DDRTree' method was used for dimensionality reduction, with the maximum number of components set to 2. The 'CellChat' [24] (version 1.6.1) package was used to infer the potential cell-cell interactions among distinct cell types. Analyses included secreted signaling, extracellular matrix-receptor interactions, and cell-cell contracts using the 'CellChatDB.human' database.

Digital cytometry

CIBERSOFTX [25] was a widely used and highly effective tool for inferring cell abundance in tissues without physical isolation. The single-cell expression matrix of two types of $\gamma\delta$ T cells (V δ 1 and V δ 2 T cells) was used as the reference matrix. The cell abundance of each sample in TCGA-BRCA and METABRIC datasets was calculated based on the corresponding bulk RNA matrix. The signature matrix was set with the following parameters: (1) Minimum Expression was set as 0.1; (2) replicates were set as 5; (3) the sampling was set as 0.5. Additionally, the absolute mode was selected to estimate cell fractions.

Differential-gene expression and enrichment analyses

The 'limma' [26] package analyzed differential gene expression between patients with high- and low- $\gamma\delta$ T cell levels. The adjusted $p < 0.05$ was considered significant. Gene sets, including 'genesets.v2023.1.hallmark.Hs' and 'c5.go.bp.v2023.2.Hs.symbols' were downloaded from the Gene Set Enrichment Analysis (GSEA) database (<https://www.gsea-msigdb.org/gsea/index.jsp>). Gene set enrichment analyses were performed by the 'clusterProfiler' [27] package (version 4.6.2) based on the hallmark and gene ontology sets. The results were visualized with the 'GseaVis' (<https://github.com/junjunlab/GseaVis>) package.

Survival analysis and in Silico Estimation of immune response to immunotherapy

The 'survminer' package analyzed the prognostic value. The Log-rank test was used to evaluate the statistical significance of survival probability across distinct groups. A $p < 0.05$ was considered significant. The 'easier' package (version 1.10.0) predicted the potential immune response to immune checkpoint inhibitors (ICIs) between high- and low- $\gamma\delta$ T cell groups. A $\gamma\delta$ T cell signature was constructed using the top 20 markers compared to other cell types. The signature score was calculated using the 'GSVA' [28] package (version 1.46.0). Pearson correlation analysis was used to estimate the correlation between the signature score and ICIs.

Analysis of single nucleotide variants and copy number variants

Masked Somatic Mutation profiles were analyzed using the 'maftools' [29] package (version 2.14.0). The Fisher exact test was used to identify significant mutant genes between high- and low- $\gamma\delta$ T cell groups. Masked Copy Number Segment data were uploaded to GISTIC2 (<https://cloud.genepattern.org/>) for analysis using online default settings. The SNP6 GRCh38 remapped probeset file was downloaded from the GDC database (<https://gdc.cancer.gov/about-data/gdc-data-processing/gdc-reference-files>) for marker files. The copy number variants (CNV) and single nucleotide variants (SNV) results were visualized using the 'maftools' package.

Radiomics model establishment

A total of 11 triple-negative breast cancers (TCGA-BRCA) with both molecular subtype information and radiomic imaging profiles were included in the study. To reduce bias caused by machine parameters, only standard double breast coil on a 1.5T GE whole-body MRI systems were used with T1-weighted DCE-MRIs as the study focus. The radiomics extension in the three-dimensional (3D) slicer was used to extract features from DCE-T1 data following the Image Biomarker Standardization Initiative guidelines. Two experienced breast surgeons with over five years of experience in breast oncology performed fully manual segmentation of the tumors, with any disagreements being confirmed by a breast radiologist with more than 20 years of experience. Eight hundred fifty-one features were obtained, including 'first-order,' 'Gray Level Co-occurrence Matrix,' 'Gray Level Dependence Matrix,' 'Gray Level Run Length Matrix,' 'Gray Level Run Length Matrix,' 'Gray Level Run Length Matrix,' 'shape,' and wavelet-based features. The resampled voxel size was set as 1,1,1. All features underwent Z-score normalization, and Pearson correlation was used to filter candidate features correlated with $\gamma\delta$ T cell infiltration ($p < 0.05$). The Elastic net regression algorithm ($\alpha = 0.5$, $n\text{folds} = 10$) was performed to establish a linear regression model for estimating the $\gamma\delta$ T cell abundance using the 'glmnet' package (version 4.1–7.0). Pearson correlation was conducted to the correlation between abundance of $\gamma\delta$ T cells and radiomic score. Receiver operating characteristic (ROC) curves were generated to assess the discriminative efficiency in identifying patients with high $\gamma\delta$ T cells.

Statistics

All the statistical methods in this study are detailed in each section of the methods chapter and were performed using R software (version 4.3.0).

RESULTS

Infiltration of $\gamma\delta$ T cells contributes to a favorable prognosis in TNBC

The overall design of this study is illustrated in Fig. 1 to preliminarily explore the presence of $\gamma\delta$ T cells in breast cancer and their relationship with patient prognosis. High-quality single-cell expression profiles were needed to infer the abundance of $\gamma\delta$ T cells in breast cancer tissues using bulk RNA profiles. First, the single-cell RNA profiles of sorted $\gamma\delta$ T cells were analyzed, and high-quality cells were obtained after removing potential doublets (Supplementary File: Figure S1). A total of 8,107 $\gamma\delta$ T cells were further divided into V δ T1 and V δ T2 cells (Figs. 2A–B). Concretely, $\gamma\delta$ T cells indicated high expression of *CD3D*, *TRDC*, *TRGC1*, and *TRGC2*, with almost no expression of *CD4* (Fig. 2C). The single-cell RNA reference matrix consisting of V δ T1 and V δ T2 cells was then used to calculate the abundance of $\gamma\delta$ T cells in TCGA-BRCA samples using CIBERSORTX.

Subsequently, the patients of TCGA-BRCA were divided into three molecular subtypes based on estrogen receptor (*ER*), progesterone receptor (*PR*), and Human Epidermal Growth Factor Receptor 2 (*HER2*) status, and patients without prognostic or survival information were excluded (Fig. 2D). Interestingly, a high abundance of $\gamma\delta$ T cells was associated with a favorable prognosis in patients with TNBC (Fig. 2G) but not significant in hormone receptor (*HR*)-positive ($p = 0.39$, Fig. 2E) and *HER2* positive patients ($p = 0.79$, Fig. 2F). These findings suggested that $\gamma\delta$ T cells may play an antitumor role in TNBC. IHC was performed to examine $\gamma\delta$ T cell infiltration in TNBC specimens to further confirm the bioinformatics. As indicated in Fig. 1H, $\gamma\delta$ T cells were observed infiltrating the tumor tissues. The $\gamma\delta$ T cells in each specimen were counted, and 61 samples were divided into high $\gamma\delta$ T ($n = 31$) and low $\gamma\delta$ T ($n = 30$) groups based on the median value (26/HPE, Fig. 2I). Consistently, high $\gamma\delta$ T cell infiltration was associated with a favorable prognosis in patients with TNBC ($p = 0.026$, Fig. 2J). Overall, our results preliminarily revealed the distinct prognostic effects of $\gamma\delta$ T cells in different breast cancer subtypes. The heterogeneous infiltration of $\gamma\delta$ T cells in TNBC tissues contributes to particular outcomes, with high $\gamma\delta$ T cell infiltration strongly associated with a favorable prognosis in TNBC.

Mutational landscapes correlated to infiltration of $\gamma\delta$ T cells

The above results provide preliminary evidence suggesting the prognostic role of $\gamma\delta$ T cells. Next, we sought to explore the factors contributing to the heterogeneity of their infiltration. Interestingly, we observed that clinical characteristics, including age ($p = 0.296$), tumor stage ($p = 0.215$), T-stage ($p = 0.274$) and race ($p = 0.766$), showed no statistically significant correlation with the

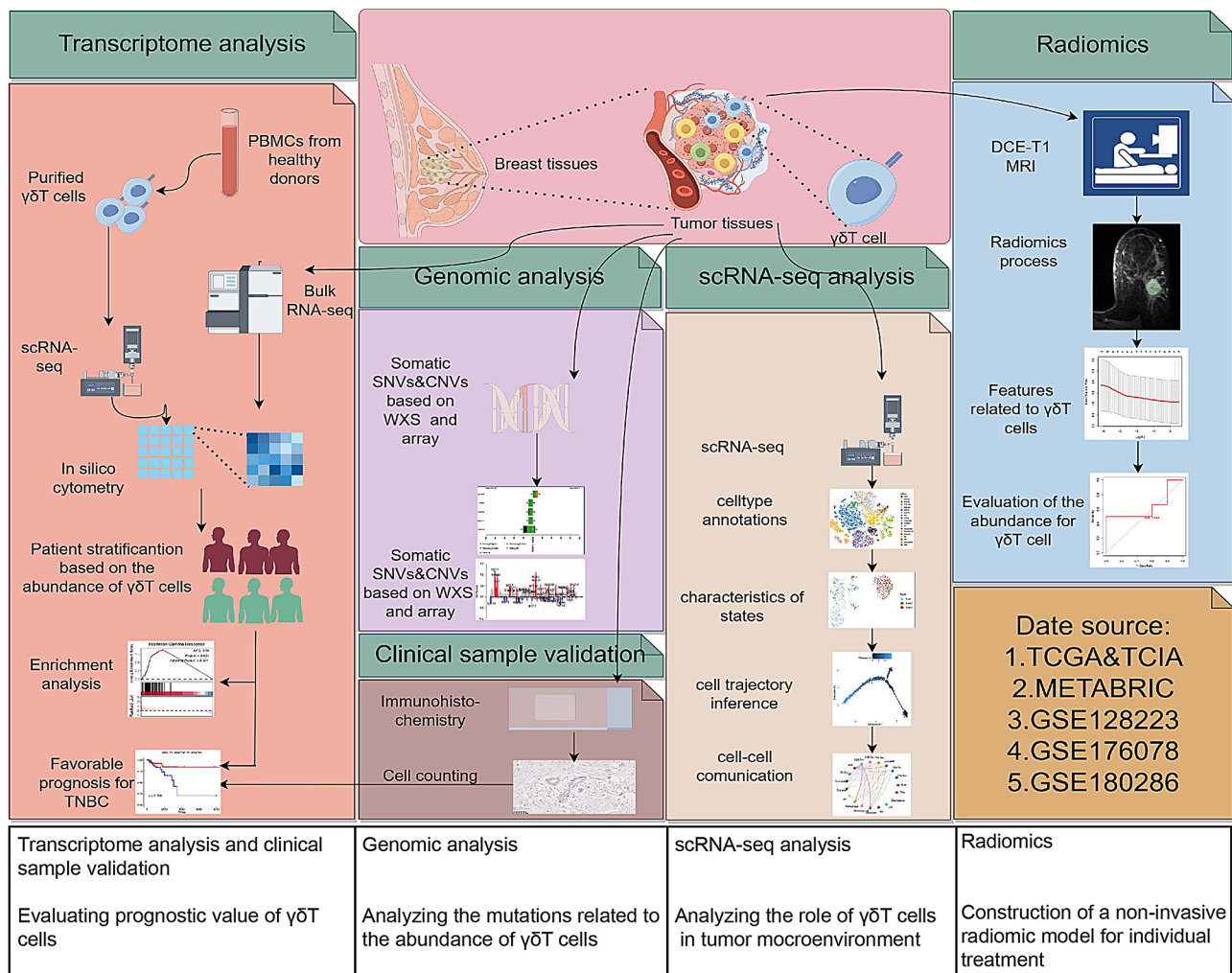


Fig. 1 Flowchart of the study design. CNVs, copy-number variants; DCE-T1, T1-weighted Dynamic Contrast-Enhanced; GSE, Gene Expression Omnibus series; METABRIC, Molecular Taxonomy of Breast Cancer International; Consortium; PBMCs, peripheral blood mononuclear cells; scRNA-seq, single-cell RNA sequencing; SNVs, single nucleotide variants; TCGA, The Cancer Genome Atlas; TCIA, The Cancer Imaging Archive; TNBC, triple-negative breast cancer; WXS, whole exome sequencing

abundance of $\gamma\delta$ T cells (Supplementary File: Table S3). Somatic gene mutations were closely related to the degree of immune cell infiltration [6, 30]. To determine whether the level of $\gamma\delta$ T cell infiltration was influenced by somatic mutations of tumor cells, patients with TNBC in the TCGA-BRCA dataset were grouped into high- and low- $\gamma\delta$ T cell infiltration groups based on the median value, consistent with the above analysis. Fisher exact test was used to identify the significant differences in somatic gene mutations between the groups. As displayed in Figs. 3A-B, genes such as MUC16, DOCK11, LAMA2, and RYR1 experienced higher variation frequencies ($p < 0.05$) in the high group ($n = 54$). Inversely, DNAH7 exhibited higher variation frequencies ($p < 0.05$) in the low group. These results suggested that the somatic mutations of MUC16, DOCK11, LAMA2, and RYR1 might be associated with $\gamma\delta$ T cell infiltration.

Interestingly, the co-occurring mutations were more prevalent in the high $\gamma\delta$ T cell group (Fig. 3D) compared to the low infiltration group (Fig. 3C), indicating a potential link between co-mutations and $\gamma\delta$ T cell infiltration. Overall, the CNV landscapes of the two groups were similar, but deletions at 4q34.3 and 13q12.12 were common in the low $\gamma\delta$ T cell group (Fig. 3E), while amplifications at 1q21.3 and 1q23.3 were more frequent in the high $\gamma\delta$ T cell group (Fig. 3F). These results imply the nuanced link between $\gamma\delta$ T cell infiltration and somatic mutations and CNVs.

$\gamma\delta$ T cells improve the TNBC microenvironment by activating immune-related pathways

Our analyses indicated that $\gamma\delta$ T cells exhibited favorable prognostic value in patients with TNBC. To further investigate the underlying mechanism, we conducted

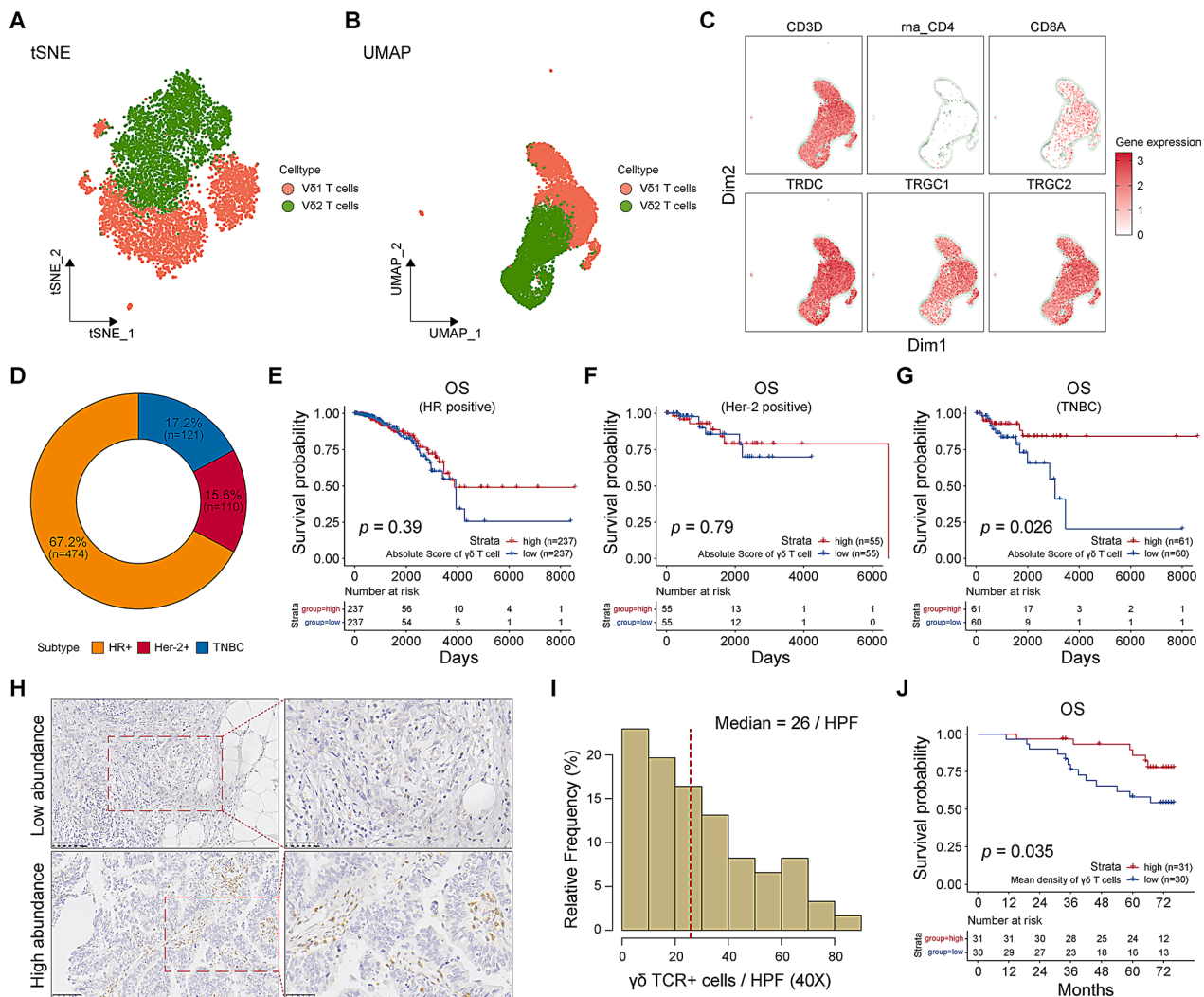


Fig. 2 Levels of $\gamma\delta$ T-cell infiltration in tumor tissue correlate with overall good prognosis in patients with TNBC. **(A–B)** T-distributed stochastic neighbor embedding (t-SNE, left) and uniform manifold approximation and projection (UMAP, right) plots of the $\gamma\delta$ T-cell reference dataset, colored by $\gamma\delta 1/\gamma\delta 2$ T-cell subtypes. **(C)** UMAP plots indicating the expression of typical $\alpha\beta/\gamma\delta$ T cells marker genes. **(D)** Pie chart demonstrating the sample size of distinct breast cancer subtypes. **(E–G)** Kaplan–Meier curves depicting the overall survival (OS) of patients with high or low $\gamma\delta$ T-cell infiltration absolute scores in the HR-positive, HER2-positive, and TNBC. **(H)** Immunohistochemical staining plots illustrate the presence of a high or low abundance of $\gamma\delta$ TCR+ cells in the 20 $\times$ and 40 $\times$ field of view. **(I)** Bar graph showing the frequency distribution of the number of $\gamma\delta$ TCR+ cells per 40 $\times$ HPF; HPF, high-power field. **(J)** Kaplan–Meier curves revealing the overall survival of patients with TNBC with high or low $\gamma\delta$ T-cell infiltration mean density—HR, hormone receptor; HER2, human epidermal growth factor 2; TNBC, triple-negative breast cancer

differential gene expression between the high $\gamma\delta$ T cell group ($n=60$) and the low $\gamma\delta$ T cell group ($n=61$). All significant genes (adjusted $p < 0.05$) are indicated in Fig. 4A. Based on these genes, we ranked them according to the fold change in expression to identify pathways potentially affected by the $\gamma\delta$ T cell abundance. The top 20 results of GSEA reviewed that immune-activation pathways, including defense response, innate immune response, positive regulation of the immune system process, and adaptive immune response, were significantly upregulated in the high $\gamma\delta$ T cell group (Fig. 4B, Supplementary File: Table S4). Furthermore, hallmark enrichment analysis revealed that pathways correlated to malignant

characteristics were downregulated, while antitumor or immune-activation pathways were widely upregulated in the high $\gamma\delta$ T cell group (Figs. 4C–J). Specifically, pathways such as TNF- α signaling (NES = 2.35, adjusted $p < 0.001$), INF- α (NES = 3.09, adjusted $p < 0.001$), allograft rejection (NES = 3.12, adjusted $p < 0.001$), JAK-STAT signaling (NES = 2.54, adjusted $p < 0.001$), INF- γ (NES = 3.59, adjusted $p < 0.001$), inflammatory response (NES = 3.11, adjusted $p < 0.001$), and complement (NES = 2.92, adjusted $p < 0.001$) were upregulated. Oxidative phosphorylation, known to be a tumor-promoting pathway in TNBC [31], was significantly downregulated (NES = -1.71, adjusted $p < 0.001$). These findings highlighted that

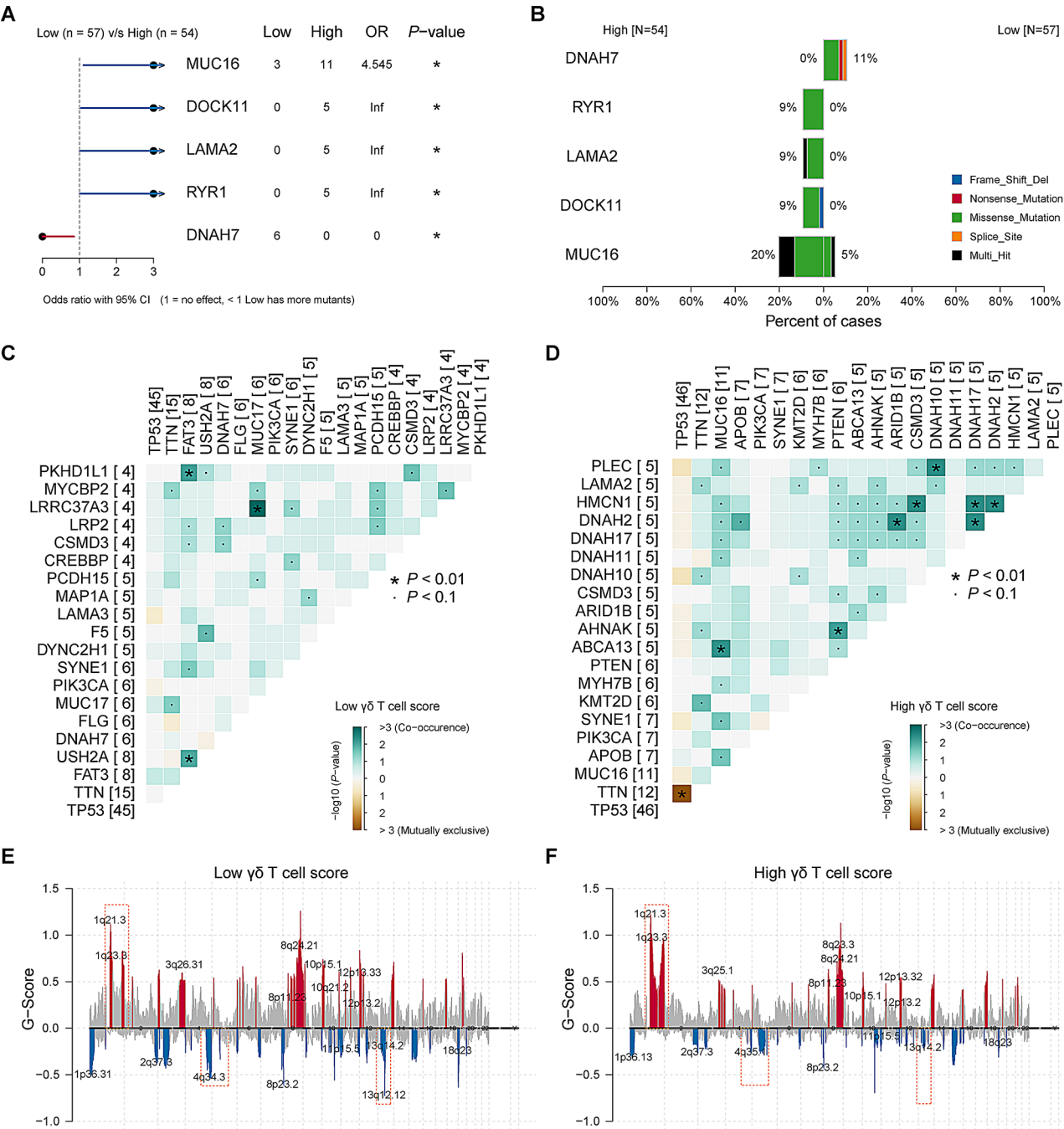


Fig. 3 The focal somatic mutations were associated with the infiltration level of $\gamma\delta$ T cells. **(A)** Forest plot illustrating the focal somatic single nucleotide variant (SNV) and copy-number alteration (CNA) significantly different between patients with high and low $\gamma\delta$ T-cell infiltrations. OR > 1 indicates a higher SNV frequency in the high $\gamma\delta$ T-cell infiltration group. OR < 1 indicates a higher SNV frequency in the low infiltration group. OR = 1 indicates consistent SNV frequency in both groups. OR, odds ratio. **(B)** Bar diagram indicating the frequency of SNVs and CNAs in the genes highlighted in panel A across the two groups. **(C-D)** Landscape of gene co-mutations in patients with low and high abundance of $\gamma\delta$ T cells. **(E-F)** visualization of CNAs in patients, stratified by low or high $\gamma\delta$ T cell infiltration levels within chromosomal context

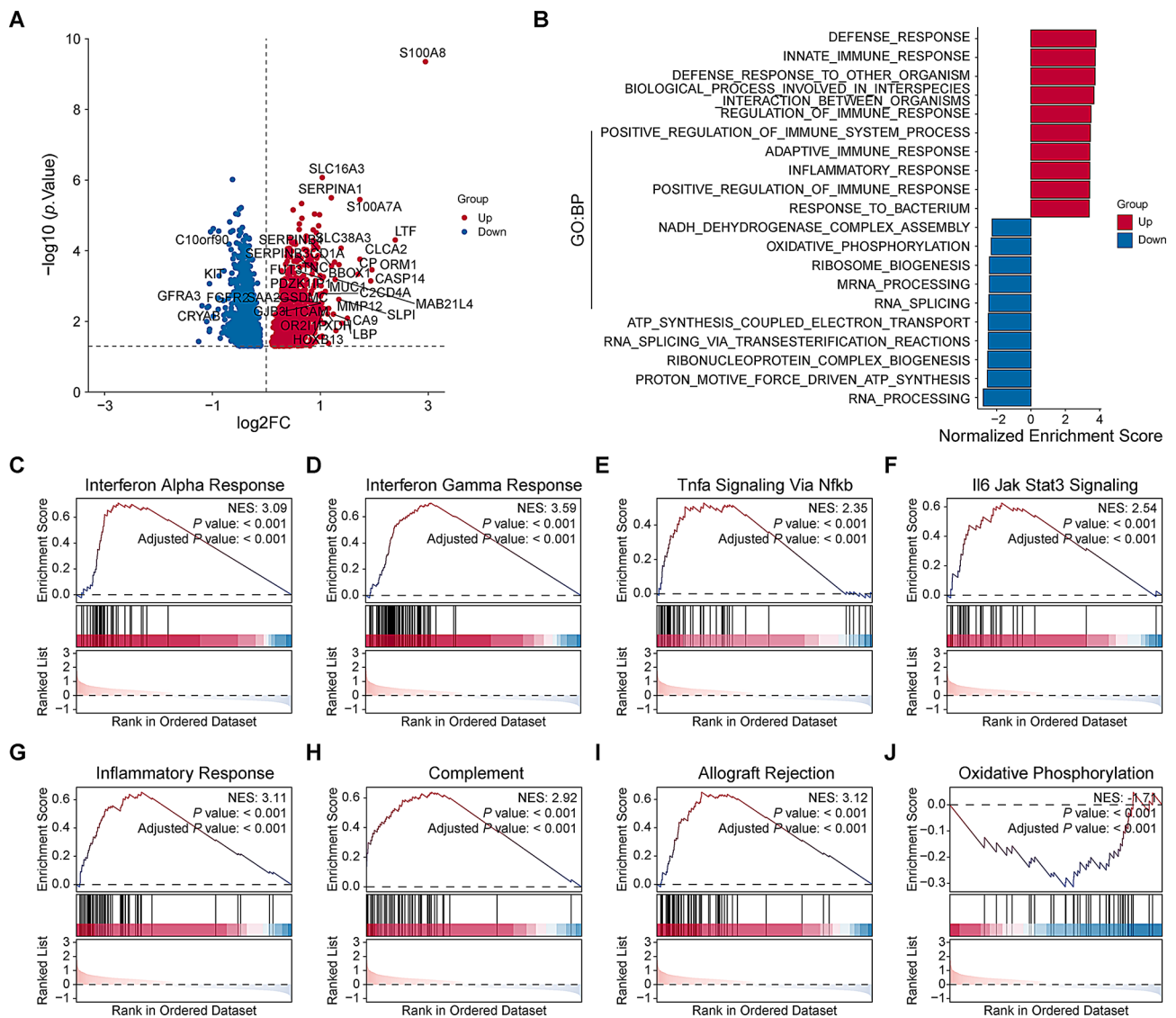


Fig. 4 Infiltration of $\gamma\delta$ T cells correlates with the activation of antitumor immunity and suppression of tumor development. **(A)** The volcano map displays differential genes between the high- and low- $\gamma\delta$ T cell groups, highlighting the most significantly upregulated or downregulated genes. **(B)** The top 10 upregulated and downregulated pathways in patients with a high $\gamma\delta$ T cell abundance were identified by gene set enrichment analysis (GSEA) based on gene ontology. **(C–J)** GSEA plots revealing representative immune-related **(C, interferon- α response; D, interferon- γ response; E, TNF signaling through NF κ B; F, IL6-JAK-STAT3 signaling; G, inflammatory response; H, complement; I, allograft rejection)** and a malignancy-related pathway **(J, oxidative phosphorylation)** significantly enriched in patients with high $\gamma\delta$ T cell abundance, identified by GSEA based on hallmark gene set. NES, normalized enrichment score

$\gamma\delta$ T cells might activate the immune system and antitumor in the TNBC microenvironment.

Intrinsic lineage evolution of $\gamma\delta$ T cells

The above studies have suggested the antitumor role of $\gamma\delta$ T cells in TNBC. To further elucidate this mechanism, it is essential to analyze the in vivo tumor microenvironment. We obtained 46,318 cells from 12 primary patients with TNBC following standard single-cell RNA preprocessing (Supplementary File: Figure S2). The cells were initially categorized into B cells, T cells, endothelial cells, epithelial cells, myeloid cells, plasmablasts, mesenchymal

cells, and cycling cells. Myeloid cells were further subdivided into macrophages, cDCs, plasmacytoid dendritic cells, and monocytes. T cells were also further classified. Ultimately, we identified and visualized 19 distinct cell types using nonlinear dimensionality reduction (Figs. 5A–B), successfully identifying $\gamma\delta$ T cells. The markers used to annotate these cell types are indicated in Fig. 5C.

The expression profiles of $\gamma\delta$ T cells were extracted to explore their potential intrinsic lineage evolution. Pseudo-time analysis suggested three different states of $\gamma\delta$ T cells in primary TNBC (Fig. 6A). $\gamma\delta$ T cells of state 3 were located at the origin of the cell trajectory, while

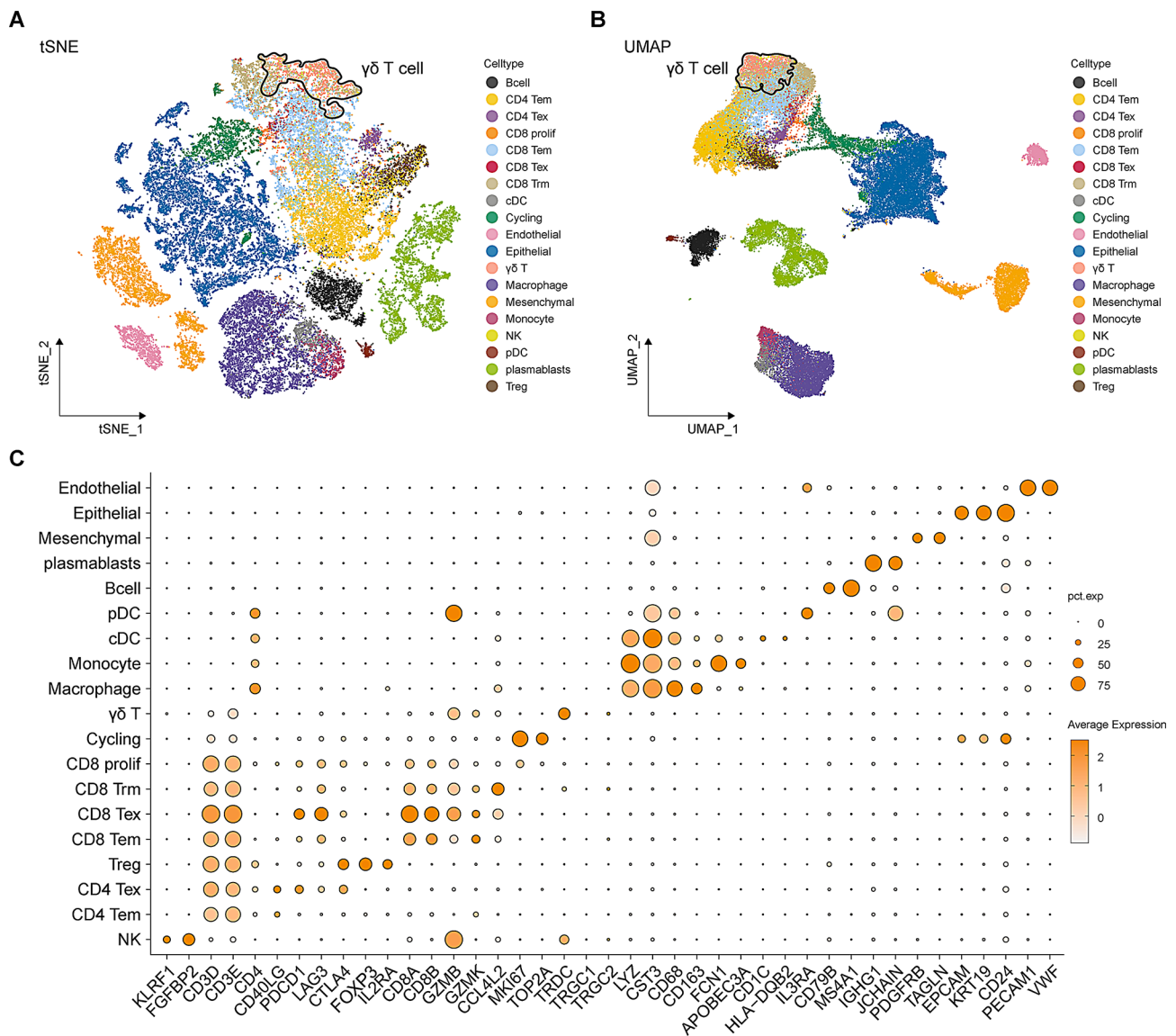


Fig. 5 Annotation of cell types in triple-negative breast cancer single-cell data. Reduced-dimensional visualization of t-distributed stochastic neighbor embedding (A) and uniform manifold approximation and projection (B) of high-quality cells in single-cell RNA datasets, with each color representing a different cell type. (C) Dot plots indicate the expression of level marker genes for each identified cell types

those in $\gamma\delta$ T cells from state 1 were at the end (Figs. 6B–C). This indicated a potential evolution of $\gamma\delta$ T cells from state 3 to 1. Further analysis of the dynamic gene revolution across pseudotime reviewed that $\gamma\delta$ T cells in the initial state exhibited high expression of *IL7R* and *TAOK* and low expression of *GNLY* and *CCL4* (Fig. 6D). As $\gamma\delta$ T cells transitioned and reached the terminal states, anti-tumor gene expression (*GNLY* and *IFNG*) increased. We examined the expression changes in five key anti-tumor genes to validate these enhanced anti-tumor effect activities. As indicated in Fig. 6E, *GNLY*, *GZMB*, *NKG7*, *XCL1*, and *XCL2* markedly increased during lineage evolution. The developmental timing of $\gamma\delta$ T cells, verified using the ‘Vector’ package) was consistent with these results

(Fig. 6F). In summary, our findings revealed the intrinsic evolution of $\gamma\delta$ T cells in TNBC and their role in generating anti-tumor effects, providing a theoretical basis for TNBC.

The interactions between $\gamma\delta$ T cells and other immune cells

Cell-cell interactions were crucial in shaping the tumor microenvironment [32]. Therefore, we analyzed the interactions between $\gamma\delta$ T cells and other immune cells to validate our previous findings. As indicated in Fig. 6G, the INF- γ signaling was observed between $\gamma\delta$ T cells and APCs (macrophages, cDCs), suggesting that $\gamma\delta$ T cells may enhance antigen presentation efficiency by promoting APC maturation and activation. The INF- γ signaling

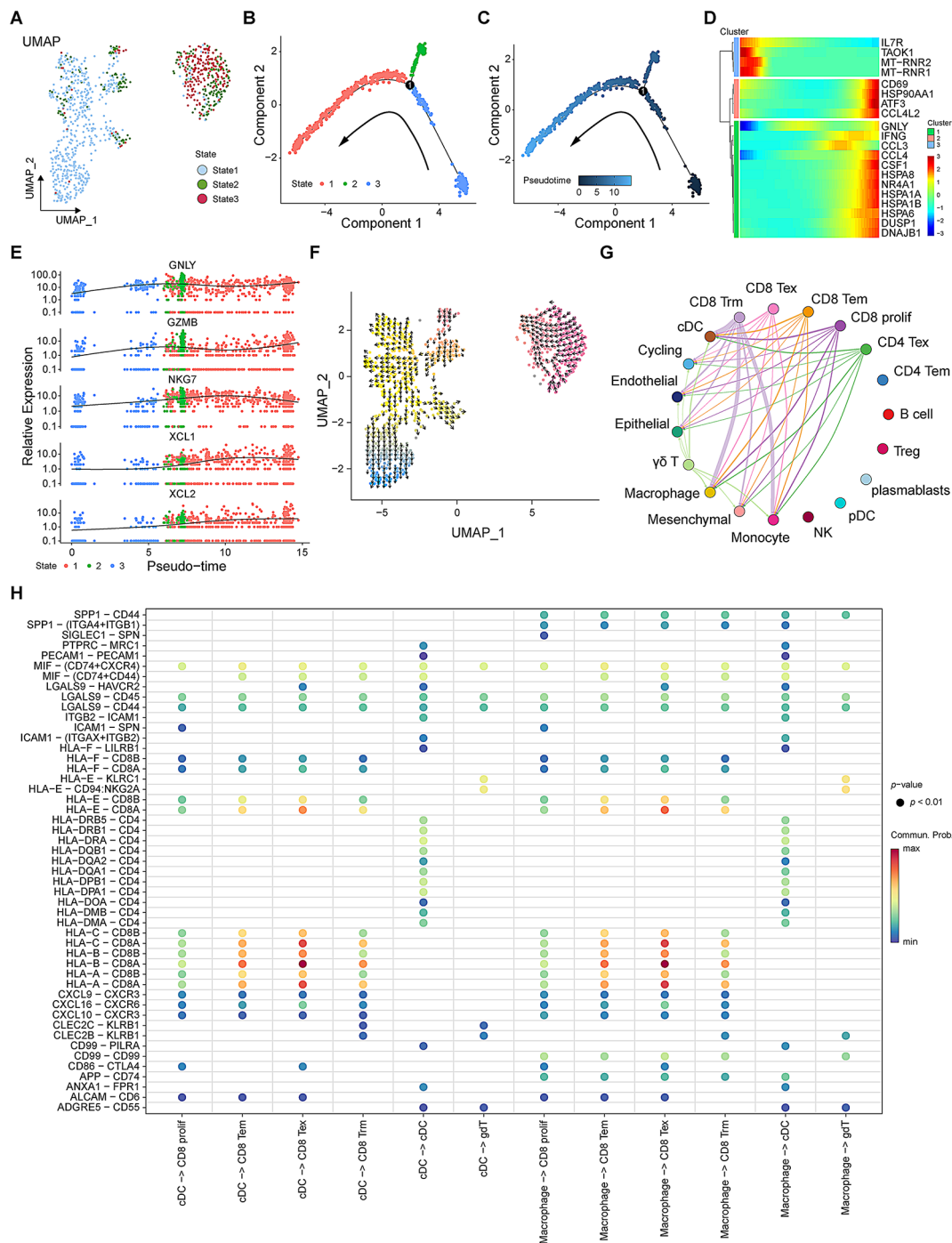


Fig. 6 Intrinsic dynamic evolution of $\gamma\delta$ T cells and their intercellular interactions within the tumor microenvironment of TNBC. **(A)** A uniform manifold approximation and projection (UMAP) plot revealing the distribution of $\gamma\delta$ T cells across three states. **(B–C)** Pseudo-time analysis of $\gamma\delta$ T cells with each color in panel B indicating a cell state and the color shade representing the development time of cell trajectory. **(D)** Heatmap of smooth expression values for the top 20 genes with the most significant expression changes in $\gamma\delta$ T cells along the evolutionary trajectory of panel C. **(E)** Dynamic gene expression of cytotoxic genes over the developmental timeline of cell trajectory. **(F)** Vector representation of developmental directions for $\gamma\delta$ T cells. **(G)** Circle plot demonstrating the communication of INF- γ signaling pathway among distinct cell subpopulations. **(H)** Dot plot illustrating the interactions between cDCs or macrophages with CD8+ $\alpha\beta$ T cells and $\gamma\delta$ T cells

to APCs also generated from CD8 T cells, indicating a possible synergistic effect between CD8 T cells and $\gamma\delta$ T cells. Additionally, non-classical MHC pathways (*HLA-E - KLRC1*; *HLA-E - CD94:NKG2A*) from macrophages or cDCs to $\gamma\delta$ T cells were observed. These results suggest that the antitumor effects of $\gamma\delta$ T cells in the TNBC microenvironment could be mediated by indirect effects with other immune cells in addition to direct cytotoxicity.

Clinical significance of $\gamma\delta$ T cells

We have demonstrated the prognostic and potential antitumor mechanisms of $\gamma\delta$ T cells; subsequently, we explored their clinical relevance. Patients with TNBC of the METABRIC cohort were divided into three groups (Supplementary File: Table S5): (1) Patients who only received chemotherapy ($n=24$); (2) patients who only received radiotherapy ($n=82$); (3) patients who received both chemotherapy and radiotherapy ($n=141$). Patients without expression profiles or complete clinical information were excluded. The abundance of $\gamma\delta$ T cells in each patient was inferred by CIBERSOFTX as previously described and divided into high and low groups by the median value. We found that a high abundance of $\gamma\delta$ T cells was associated with a favorable overall survival in patients who received only radiotherapy ($p=0.021$, Fig. 7B). A suggestive association was also observed

between the $\gamma\delta$ T cells abundance and OS in patients who received chemotherapy ($p=0.056$, Fig. 7A). Interestingly, no significant correlation was found between $\gamma\delta$ T cell abundance and overall survival (OS) in patients who received both treatments ($p=0.15$, Fig. 7C). These findings suggested that $\gamma\delta$ T cell infiltration might have important clinical significance in guiding the postoperative treatment for patients with TNBC.

Patients with a high abundance of $\gamma\delta$ T cells May benefit from immunotherapy

Immunotherapy has recently emerged as a potential treatment for TNBC, but only a few patients respond to it [33, 34]. Consequently, we aimed to investigate whether the abundance of $\gamma\delta$ T cells could predict immunotherapy response. Using the 'Findallmarker' function in the Seurat package, we identified the top 20 markers ($p<0.05$) of $\gamma\delta$ T cells compared to other cell types. These markers were used to construct a $\gamma\delta$ T cell signature. We found that this signature was strongly associated with the expression of immune checkpoints, including PD-L1 (CD274, $R=0.73$, $p<0.001$, Fig. 7D), CTLA-4 ($R=0.76$, $p<0.001$, Fig. 7E), and LAG3 ($R=0.84$, $p<0.001$, Fig. 7F). To further investigate the immune response between the high- and low- $\gamma\delta$ T cell group in the TCGA cohort (Supplementary File: Table S6), we conducted an EaSIeR

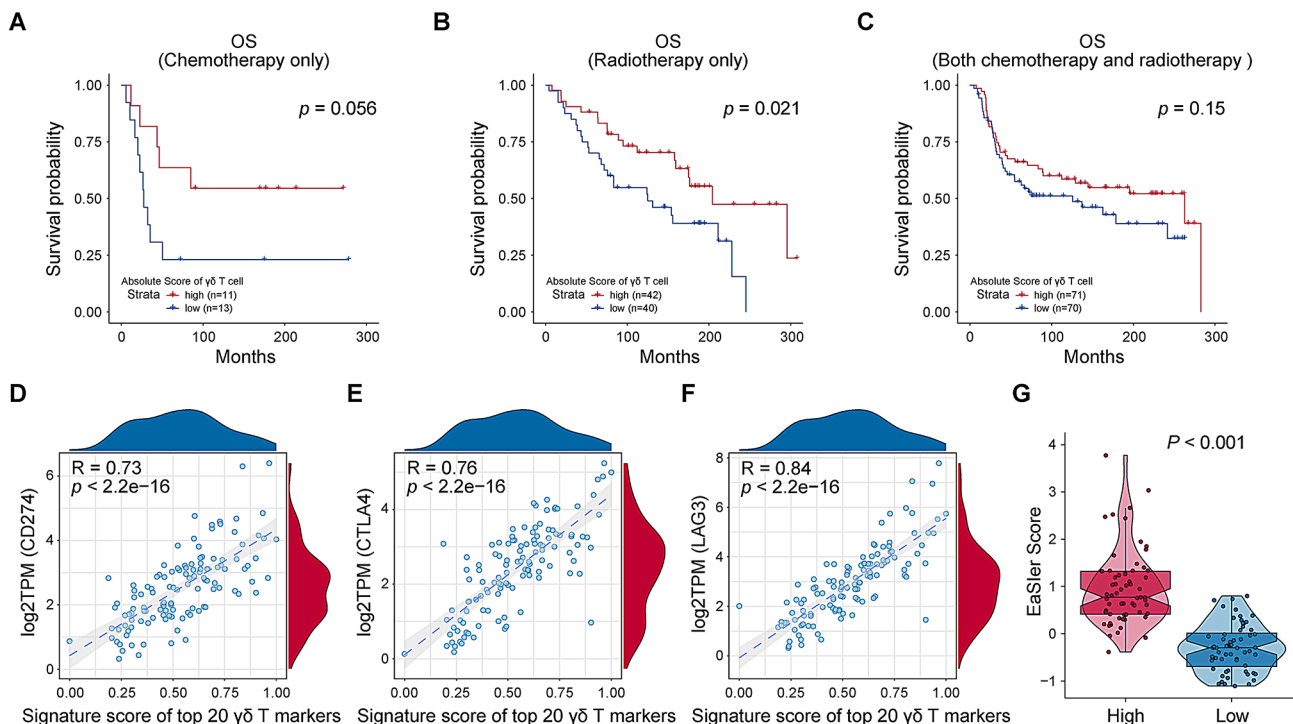


Fig. 7 The infiltration level of $\gamma\delta$ T cells could be a predictor for chemotherapy, radiotherapy as well as immunotherapy. **(A)** The Kaplan-Meier graphs revealing the overall survival rates of patients who received chemotherapy alone. **(B)** The Kaplan-Meier graphs illustrating the overall survival rates of patients who received radiotherapy alone. **(C)** Kaplan-Meier graphs demonstrating the overall survival rates of patients who received both radiotherapy and chemotherapy. The scatter plots illustrating a significant positive association between the expression level of the top 20 genes that characterize $\gamma\delta$ T cells and the expression level of CD274 **(D)**, CTLA4 **(E)**, and LAG3 **(F)**. **(G)** Boxplot of EaSIeR scores in high- and low- $\gamma\delta$ T cell group patients

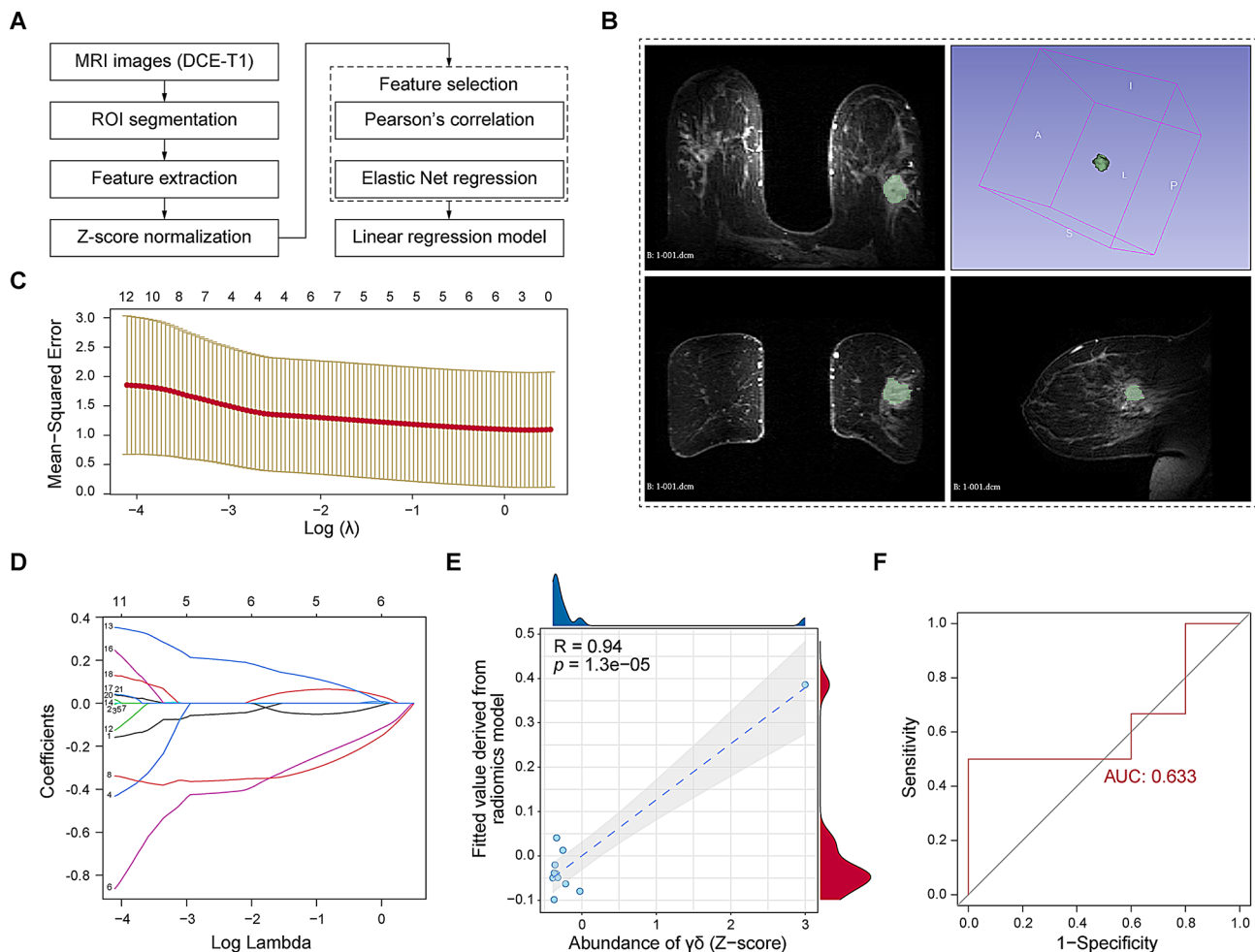


Fig. 8 A linear regression model was developed using MRI-based radiomics analysis to estimate the abundance of $\gamma\delta$ T-cells in TNBC tissue. **(A)** Schematic diagram illustrating the radiomics analysis process. **(B)** Example indicating segmentation of gross tumor volume on DCE-T1 MRI. **(C)** Parameter tuning plot for the Elastic net regression analysis. **(D)** Distribution of coefficients for variables in the Elastic net regression is presented, with each curve representing a radiomics feature filtered using Pearson's correlation. **(E)** Pearson's correlation was calculated between the z-score normalized abundance of $\gamma\delta$ T cells assessed by the transcriptome and the fitted value obtained from the linear regression radiomics model. **(F)** The ROC curve indicating the model's ability to discriminate the abundance of $\gamma\delta$ T cells

analysis. As indicated in Fig. 7G, the high $\gamma\delta$ T cell group exhibited a higher EaSIer score than the low group. These findings suggested that patients with a high abundance of $\gamma\delta$ T cells might be suitable for immune checkpoint therapy.

Noninvasive MRI radiomics could be a promising tool for evaluating the infiltration of $\gamma\delta$ T cells in TNBC

Given the clinical benefit of $\gamma\delta$ T cells, we investigated their potential translational applications. The workflow is illustrated in Fig. 8A. We included 11 patients with TNBC from the (TGCA cohort) who had undergone both bulk RNA sequencing and DCE-MRI. The radiomics plugin in 3D slicer software was used to segment the DCE-MRI images (Fig. 8B), resulting in the identification of 851 radiomics features. Pearson correlation analysis was then conducted to identify features significantly related

to the abundance of $\gamma\delta$ T cells ($p < 0.05$, Supplementary File: Table S7). An elastic net regression algorithm was used to construct a model for estimating $\gamma\delta$ T cell abundance (Figs. 8C–D). The $\gamma\delta$ T radiomic score model (Supplementary File: Table S8) was established as follows: $\gamma\delta$ T radiomic score = $-0.06639138 \times \text{wavelet-LLH.glcmm} - 0.08090734 \times \text{wavelet-HLH.firstorder.Skewness}$. A strong positive correlation was observed between the $\gamma\delta$ T cell abundance and the radiomic score ($p = 0.000013$, $R = 0.94$, Fig. 8E). Exploratory analysis based on ROC curve also showed a potential association between radiomic score and the infiltration of $\gamma\delta$ T cell infiltration (AUC = 0.633, Fig. 8F). In summary, we developed a promising radiomic model for recognizing $\gamma\delta$ T cell abundance, providing a noninvasive strategy for individual therapy in patients with TNBC.

Discussion

To our knowledge, this is the first study to comprehensively elucidate the role of $\gamma\delta$ T cells in TNBC by integrating extensive data. By combining radiomics, transcriptome, single-cell transcriptome, and genomics, the study revealed for the first time identified factors influencing $\gamma\delta$ T cell infiltration, the intrinsic evolution of $\gamma\delta$ T cells, the impact of $\gamma\delta$ T cells on chemoradiotherapy efficacy, and the relationship between $\gamma\delta$ T cell abundance and radiomic features.

The antitumor role of $\gamma\delta$ T cells in breast cancer has been controversial. Ma et al. [35] found that intratumoral $\gamma\delta$ T cells were positively correlated with FOXP3(+) cells and CD4(+) T cells but negatively correlated with CD8(+) T cells in breast cancer tissues, which was associated with poor prognosis. Another study [36] reported that a subgroup of $\gamma\delta$ T cells producing IL-17 contributed to an immune-suppressive microenvironment, leading to distant metastasis in breast cancer. In contrast, Michot et al. [37] demonstrated that the infiltration level of $\gamma\delta$ T cells was associated with favorable OS and RFS in TNBC. These contradictory findings may stem from the high heterogeneity of breast cancer. Therefore, multi-omics profiling should be integrated to explore the role of $\gamma\delta$ T cells in different molecular subtypes (HR, HER2, TNBC) of breast cancer to elucidate their prognostic role. Our results confirmed that the high level of $\gamma\delta$ T cell infiltration was associated with a favorable prognosis ($p=0.026$) in TNBC, while no significant associations were observed in patients with HR+ or HER2+. IHC further supported our bioinformatics findings, suggesting that the heterogeneity of $\gamma\delta$ T cells in the microenvironment of different breast cancer subtypes contributes to their distinct clinical outcomes. IFN- γ [38] is a cytokine with a pro-inflammatory effect, coordinating the innate and adaptive immune response and inhibiting cell proliferation while including cell death in non-immune cells. We discovered that immune-activated signaling pathways, including IFN- γ , TNF- α , and inflammatory response, were significantly upregulated in the high $\gamma\delta$ T cell group, suggesting that $\gamma\delta$ T cells improved the TNBC tumor microenvironment and activated the immune system. Furthermore, IFN- γ signaling was highly upregulated from $\gamma\delta$ T cells to APCs (cDCs and macrophage), indicating that the $\gamma\delta$ T cells could activate APCs and enhance antigen presentation pathway to exert an antitumor effect in TNBC. Meanwhile, $\gamma\delta$ T cells received MHC signaling from cDCs or macrophages, allowing them to exert a direct antitumor effect through the NKG2A or KLRC1 receptors. Importantly, the intrinsic lineage evolution of $\gamma\delta$ T cells indicated a gradual upregulation of cytotoxic markers, including GNLY, GZMB, NKG7, XCL1, and XCL2. In conclusion, these findings underscored the critical role of $\gamma\delta$ T cells in the TNBC microenvironment and provide a

novel strategy and theoretical basis for personalized therapy targeting $\gamma\delta$ T cells in TNBC.

The infiltration level of immune cells is influenced by many factors, and the heterogeneity of infiltration has been indicated to lead to varying patient prognosis and therapy responses [39, 40]. Our results indicate that a high infiltration of $\gamma\delta$ T cells in TNBC is closely associated with somatic mutations in *MUC16*, *RYR1*, *LAMA2*, and *DOCK11*. Interestingly, *MUC16* (CA125) is a well-known biomarker in multiple cancers, including ovarian and pancreatic cancer [41, 42]. Wang et al. [43] reported that the mutation of *RYR1* is closely associated with a favorable prognosis in lung adenocarcinoma. Additionally, loss-of-function mutations in *DOCK11* have been linked to severe immune dysregulation and systemic inflammation [44]. *LAMA2* has been identified as a tumor suppressor in breast cancer [45] and is among the top 10 most frequently mutated genes in inflammatory breast cancer [46]. Although the correlation between these gene mutations and $\gamma\delta$ T cell infiltration requires further experimental validation, these studies provide preliminary evidence of their potential significance. Overall, our findings reveal previously unreported somatic mutations associated with $\gamma\delta$ T cell infiltration, which may serve as indicators of $\gamma\delta$ T cell abundance.

Given the drug resistance, heterogeneity, and malignancy of TNBC, combining multiple treatments is essential to improve patient prognosis [47]. Although chemotherapy has been a standard treatment for TNBC, only some patients achieve a complete response. Immunotherapy in TNBC faces similar challenges [48]. Therefore, new biomarkers are needed to identify specific populations benefitting from corresponding treatment. Our results revealed that patients with high $\gamma\delta$ T cell infiltration were more likely to benefit from either chemotherapy alone ($p=0.056$) or radiotherapy alone ($p=0.021$). However, no significant difference was observed in patients receiving chemotherapy and radiotherapy ($p=0.15$). Interestingly, patients with high $\gamma\delta$ T cell infiltration exhibited a higher EaSIer score, suggesting they might benefit from ICIs. In summary, $\gamma\delta$ T cell abundance could be a promising predictor for estimating the clinical effectiveness of chemoradiotherapy or immunotherapy.

Radiomics, which enables the extraction of imaging data to support clinical decision-making, provides a noninvasive method for predicting the diagnosis and prognosis of diseases [49]. For instance, Wang et al. [50] developed a high-quality prognostic model for colorectal lung metastasis patients based on radiomics derived from histopathological images. Aoude et al. [51] confirmed that radiomics features from computed tomographic (CT) images could reflect a correlation with the tumor microenvironment. Additionally, MRI-based radiomics

has been used to estimate the abundance of tumor-infiltrating lymphocytes [52]. In this study, we constructed a radiomics model to estimate the abundance of $\gamma\delta$ T cells. Due to the limited availability of multi-omics data, we could only collect 11 radiological profiles of TNBC derived from the TCIA database corresponding to patients with TCGA. This limited data prevented our internal and external validation, which may affect the robustness of our model. However, our results indicated that the radiomics model was strongly associated with the abundance of $\gamma\delta$ T cells ($p=0.000013$). They exhibited potential in distinguishing between patients with high and low $\gamma\delta$ T cells ($AUC=0.633$). As highlighted by previous studies, multi-omics technologies and imaging models have demonstrated the potential to improve existing clinical pathways, enhance disease diagnostic accuracy, and support personalized treatment strategies in medical practice [53–55]. Our radiomic model leverages non-invasive MRI examinations to predict the cellular composition within tumor tissues of TNBC patients during their clinical course. This approach can assist clinicians in treatment planning, follow-up care, and the selection of personalized therapeutic strategies, thereby improving the efficiency and effectiveness of clinical practice.

This study has several limitations. First, our analysis is primarily based on the reanalysis of publicly available datasets. The role of $\gamma\delta$ T cells in TNBC still requires further validation in future research. We will extract human primary $\gamma\delta$ T cells, co-culture them with TNBC cells, and analyze and verify the function of $\gamma\delta$ T cells by detecting the biological change such as apoptosis and migration of TNBC that co-cultured. In addition, we will construct an orthotopic xenograft tumor model which mix inoculate murine $\gamma\delta$ T cells and murine TNBC cells to observe the effect of $\gamma\delta$ T cells on TNBC growth. Additionally, the deconvolution algorithm used to predict $\gamma\delta$ T cell content in RNA-seq samples may introduce some biases and might not fully reflect the actual cellular composition in the samples. Encouragingly, a study comparing the performance of over ten commonly used deconvolution algorithms reported that CIBERSORTx provided results that were closer to actual values [56]. Furthermore, we identified potential interactions between $\gamma\delta$ T cells and other cell types through single-cell RNA sequencing. However, in vivo and in vitro experiments are necessary to further validate these interactions. We also developed a radiomic model to predict the level of $\gamma\delta$ T cells in TNBC tumor tissues. It is important to note that this radiomic model is exploratory and not intended to serve as a robust tool for immediate clinical application. The limited availability of imaging data may impact reliability and stability of the model. To address this, we plan to conduct multi-center studies in the future to collect more MRI data and tumor

specimens. As sequencing technologies and artificial intelligence algorithms continue to advance, some of the current limitations may be mitigated. Future multi-omics research has the potential to accelerate clinical translation, ultimately enabling more precise and effective treatment strategies for TNBC patients.

Conclusion

In summary, our study comprehensively revealed the antitumor effect of $\gamma\delta$ T cells in the TNBC microenvironment. Patients with a high abundance of $\gamma\delta$ T cells are suitable for chemotherapy or chemotherapy alone and are more likely to respond to immunotherapy. Our findings provide a novel insight and theoretical basis for personalized therapy in patients with TNBC.

Abbreviations

TNBC	Triple-negative breast cancer
TCR	T cell receptors
MHC	Major histocompatibility
RNA	Ribonucleic acid
DCE-MRI	Dynamic contrast-enhanced magnetic resonance imaging
APCs	Antigen-presenting cells
cDCs	Conventional dendritic cells
CNV	Copy number variation
SNV	Single nucleotide variation
IHC	Immunohistochemistry
TCIA	The Cancer Imaging Archive
GSEA	Gene Set Enrichment Analysis
ICIs	Immune checkpoint inhibitors
ROC	Receiver operating characteristic
ER	Estrogen receptor
PR	Progesterone receptor
HER2	Human epidermal growth factor receptor 2
HR	Hormone receptor
CT	Computed tomographic
MUC16	Mucin 16
DOCK11	Dedicator of Cytokinesis 11
LAMA2	Laminin Subunit Alpha 2
RYR1	Ryanodine Receptor 1

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12885-025-14029-8>.

Supplementary Material 1
Supplementary Material 2
Supplementary Material 3
Supplementary Material 4
Supplementary Material 5
Supplementary Material 6
Supplementary Material 7
Supplementary Material 8
Supplementary Material 9

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Author contributions

Guixin Wang, Xin Wang, Yao Tian and Yingxi Li designed the study and collected research data; Guixin Wang and Shuo Wang were responsible for the interpretation of the results. Wenbin Song, Chenglu Lu, Zhaohui Chen, Long He and Xiaoning Wang analyzed data and drafted the manuscript; Yizeng Wang, Cangchang Shi, Zhaoyi Liu, and Yue Yu provided the overall guidance. All authors read and approved the final Manuscript.

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Data availability

The datasets presented in this study can be found in online repositories. The names of the repository/repositories and accession number(s) can be found in the article.

Declarations

Ethics approval and consent to participate

The study protocol was in line with the ethical guidelines of the 1975 Declaration of Helsinki and was approved by the Ethical Committee of Tianjin Medical University Cancer Institute and Hospital. Informed consent to participate in the study was obtained from each patient that was recruited.

Consent for publication

Informed consent to publish was obtained from each patient that was recruited.

Competing interests

The authors declare no competing interests.

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